DRAFT for team review. Replace "Team##" with our team number; all numbers come from our research notes and the portal backtest of file SHA256 1f22bae1...

Team## QuantiFi 2026: A Multi-Module Market-Making and Relative-Value Trader

Summary. Our trader runs six independent modules, one per product family, inside a single `Trader.run`. Each product is owned by exactly one module, so modules never fight over inventory. On the official portal backtester our submitted file finished the three training days (-2, -1, 0) at about **327,250 coins**, with a steadily rising equity curve on all three days.

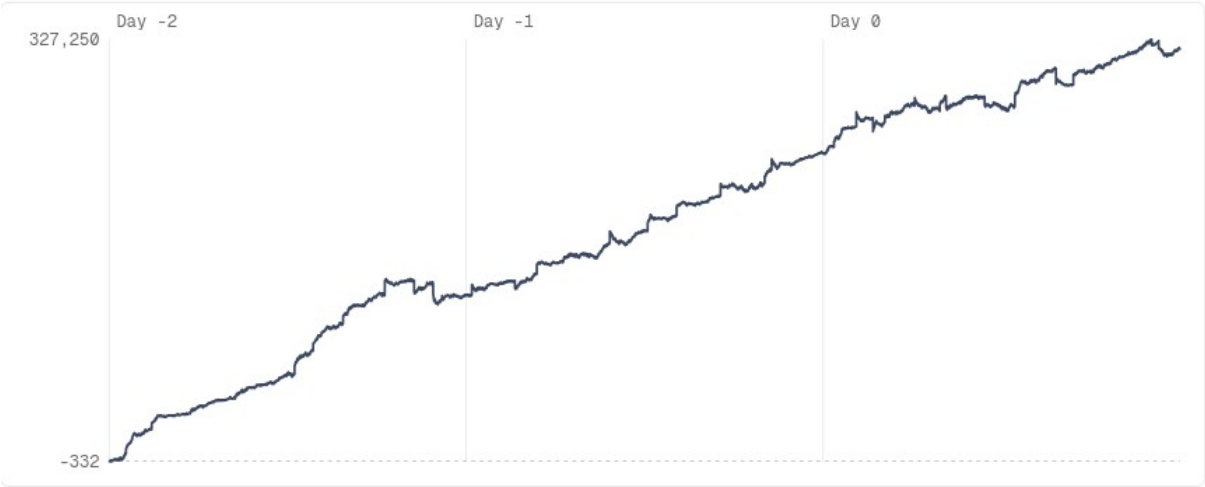


Figure 1. Official portal backtest equity across training days -2, -1 and 0 (portal screenshot, cropped).

1. Strategy modules

- **AMETHYST (market making).** Quotes around the 10,000 anchor from the official example, with position capacity raised to the official limit of 50.
- **VOUCHER options (volatility smile).** For each strike we fit a quadratic smile of total volatility against ln(K/S) using all *other* strikes (leave-one-out), and trade a strike when its quote deviates from the fitted value by more than a threshold of 7, up to 50 per strike, unhedged. With r = 0 no time-to-expiry guess is needed. The two wing strikes (6800, 8100) are used for fitting only.
- **PAPER (trend gate).** A trailing 50-observation OLS fit gives a level plus 50 times the slope as a forecast; we enter when |slope| > 0.05 and the forecast clears the tradable price, and exit when the signed slope falls to 0.02.
- **TOFFEE and JELLY (mean reversion).** Fade deviations from anchors of 5,000 (entry 40) and 2,000 (entry 80), closing back at the anchor, capacity 50 each.
- **TELLURIUM (trend).** EMA 250 / 1500 crossover, capped at the official limit of 40.

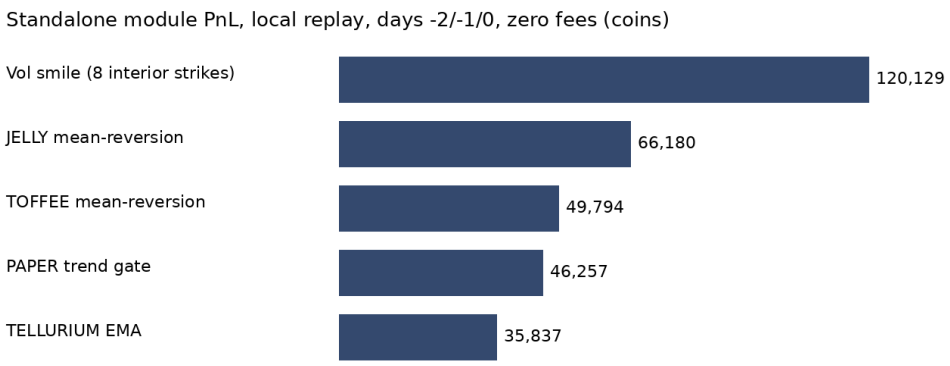


Figure 2. Stand-alone PnL of each module in our local replay engine (zero fees, days -2/-1/0). Because modules trade different products, the portfolio PnL is close to the sum of the modules.

2. Why it works

Each module targets a different, simple inefficiency: a stable fair value (AMETHYST, TOFFEE, JELLY), a cross-sectional pricing relationship (the option smile), or persistent drift (PAPER, TELLURIUM). Since the products do not overlap, drawdowns arrive at different times and partly offset. The largest single contributor is the smile module: eight interior strikes earned about 120k locally, far more than the 7500 strike alone (about 22k).

3. Engineering for correctness

- **Single owner per product.** Naively stacking two PAPER strategies lost 75,556 coins; assigning one owner removed the conflict.
- **Pending-order accounting.** Our first version only reserved room against the official limit, not against the target position, so exits could overshoot zero (for example long 33 with sell 8 pending, then sell 33). The fixed version subtracts same-side pending orders from the target gap; this turned a TOFFEE delayed-fill scenario from -253,241 to +50,542.
- **Limits.** All caps match the official per-product limits (50 for AMETHYST, PAPER, TOFFEE, JELLY and each voucher strike; 40 for TELLURIUM).
- **Performance.** Worst p99 call time locally about 0.29 ms; `traderData` at most about 1 KB.

4. Robustness checks (local engine)

Scenario	Total PnL
Base	319,295
One-observation delay	309,572.5
Combined stress (delay, strict passive matching, half passive volume)	308,565
No passive fills	318,197

Day	-2	-1	0	Total	Max intraday drawdown
Local PnL	128,797.5	109,751	80,746.5	319,295	19,868.5

Of the local total, about 288,871 is realized and 30,424 is unrealized mark-to-mid. Results are robust to execution delay, which suggests the edge does not depend on queue position.

5. Trade-offs and limitations

- All three days were used for research, so these are in-sample results; we have no true out-of-sample day.
- Excluding the two wing strikes from trading was decided after seeing their losses (-10,455 and -63), which is a selection bias.
- TOFFEE and JELLY anchors are hard-coded; a regime change would break those legs. TELLURIUM lost 3,946 on day 0 with a large own drawdown, and we kept it for return at the cost of risk. A more conservative variant (dropping TELLURIUM) earned 283,034 locally with a much smaller drawdown of 8,157.
- All option positions share LYCHEE underlying risk, so the ten strikes are not independent bets.